TEMMED-BENCH: EVALUATING TEMPORAL MEDICAL IMAGE REASONING IN VISION-LANGUAGE MODELS

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Contents

- Clinical AI vs. Real-world Practice
- TEMMED-BENCH:
  - Dataset & Tasks
- Benchmark Overview
- Retrieval Augmented Generation
  - Problem Formulation
  - Pair-wise Image Retrieval
  - Results
  - Ablation Study and Analysis
  - Top 1 Retrieval Hack
- Discussion

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Clinical AI vs Real World Practice

The Clinical Reality

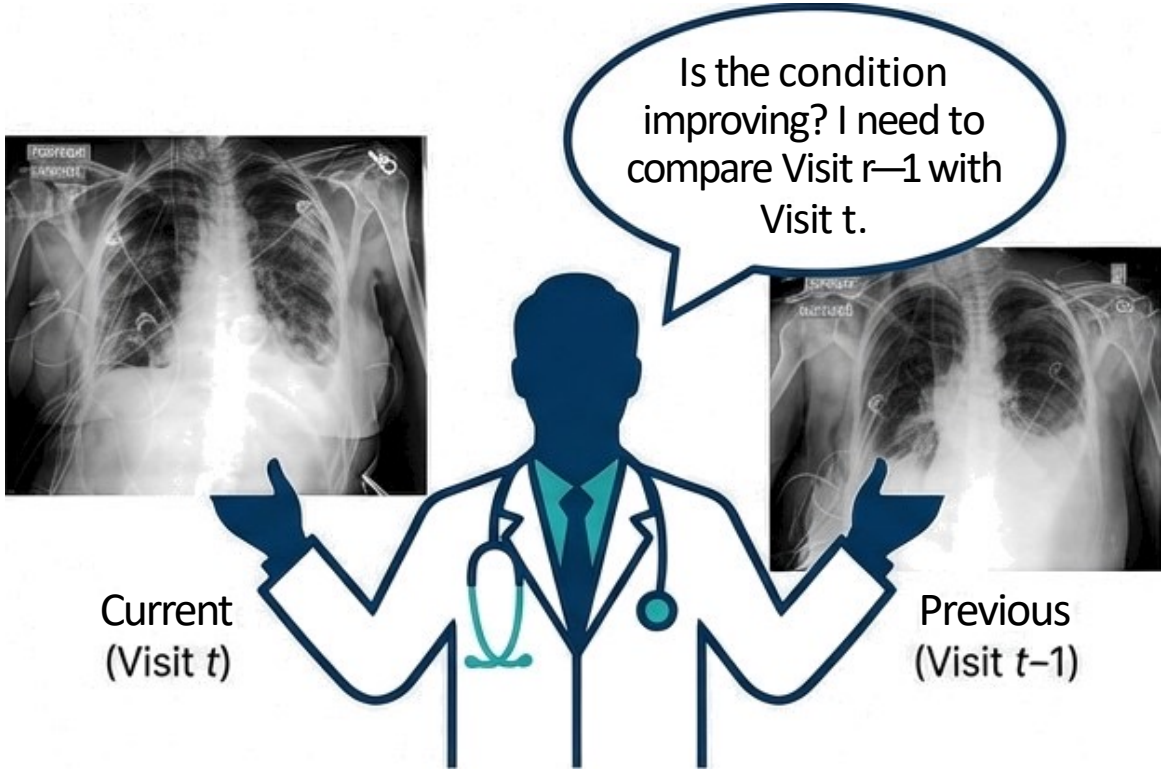

Image Source:Gemini

Doctors track changes. Diagnosis is a relative process.

The AI Status Quo

Image Source:Gemini

Benchmarks like MIMIC-CXR force models to look only at Visit t.
They recognize patterns but cannot reason about progress.

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What is TEMMED-BENCH?

First Benchmark for Temporal Reasoning

Tests if AI can reason about changes in patient conditions over time

Three Task Suite

Visual Question Answering (VQA), Report Generation, Image-Pair Selection

Multi-Modal RAG Exploration

Demonstrates effectiveness of retrieving both visual and textual medical information

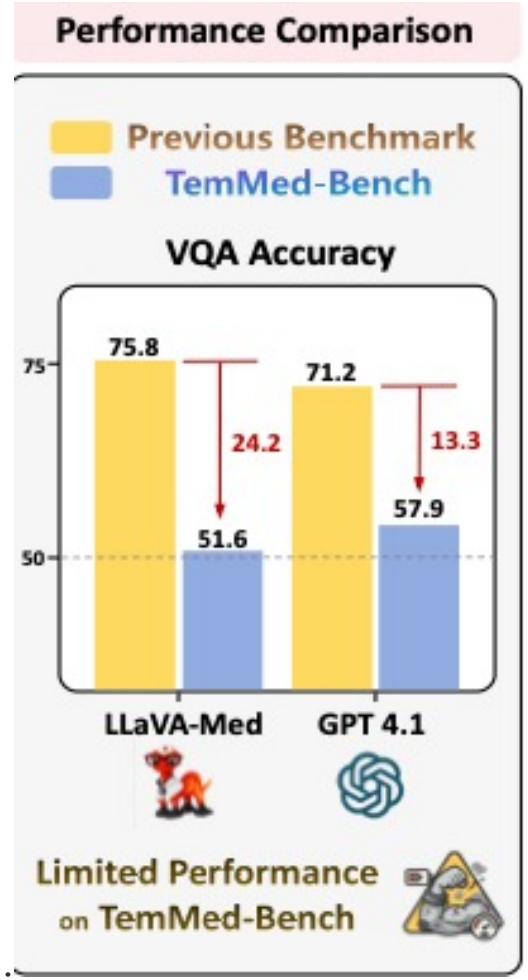


Figure 1: An example from TEMMED-BENCH. Previous benchmarks (Johnson et al., 2019; Hu et al., 2024) mainly focused on analyzing a single-visit image. However, real-world clinical practice requires doctors to monitor changes in patients' conditions over time. The previous benchmark in the rightmost chart is a VQA variant (Xia et al., 2025) of the MIMIC-CXR dataset (Johnson et al., 2019).

18,144 instances • 3 tasks • 17,144 knowledge corpus

Three Pillars of Temporal Evaluation

VQA

Past Present

Has the intestinial edema improved?

Yes /No

Report Generation

Past Present

Task: Generate a report analyzing the changes.

Interval improvement in aeration at right lung base, with decreased opacity compared to the prior study. Stable mild pulmonary edema is noted in the upper lobes, unchanged from

previous examination.

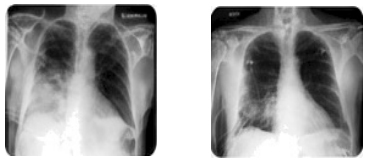
Image Pair Selection

Which option matches: "increased middle lobe consolidation"?

A

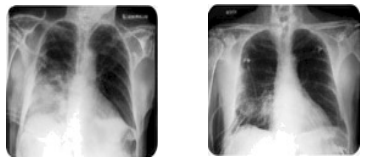
[Incorrect] ✗ persistent ... consolidation

B



[Incorrect] ✗ unchanged ... consolidation

C



[Correct!] increased middle lobe consolidation

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Related work

Benchmarks	Task	Historical Conditions	Multi-Image Input
VQA-RAD (Lau et al., 2018)	VQA	✗	✗
SLAKE (Liu et al., 2021)	VQA	✗	✗
PathVQA (He et al., 2020)	VQA	✗	✗
PMC-VQA (Zhang et al., 2024)	VQA	✗	✗
PubMedVision (Chen et al., 2024)	VQA	✗	✗
OmniMedVQA (Hu et al., 2024)	VQA	✗	✗
Harvard-FairVLMed (Luo et al., 2024)	Report	✗	✗
IU-Xray (Demner-Fushman et al., 2015)	Report	✗	✗
CheXpert Plus (Chambon et al., 2024)	Report	✗	✗
MIMIC-CXR (Johnson et al., 2019)	VQA + Report	✗	✗
TEMMED-BENCH (Ours)	**VQA+Report+Image-pair**	✓	✓

Table 1: Comparison with previous works. TEMMED-BENCH focuses on evaluating LVLMs in temporal reasoning over multiple medical images. VQA: visual question answering; Report: report generation; Image-pair: image-pair selection.

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Benchmark Overview

VQA:

An LVLM M takes a historical image i_h a current image i_c and a textual query q about condition change, and outputs a binary response: **"Yes"** or **"No."**

Report Generation:

Model M takes a historical image i_h, a current image i_c ,and a task instruction q, and generates a report analyzing condition changes between the images.

Image Pair Selection

M is given three image pairs – IA ($[i_{h1}, i_{c1}]$), IB ($[i_{h2}, i_{c2}]$), and IC ($[i_{h3}, i_{c3}]$) – along with a textual question q, and is required to output the choice of A, B, or C that best matches the medical statement in q.

Statistic	Questions	Images	Choice
VQA	2000	2	2
Report	1000	2	-
Image Pair	862	6	3
Corpus	17,144	2	-

Table 2: Key Statistics of TEMMED

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Engineering a Dataset for Change

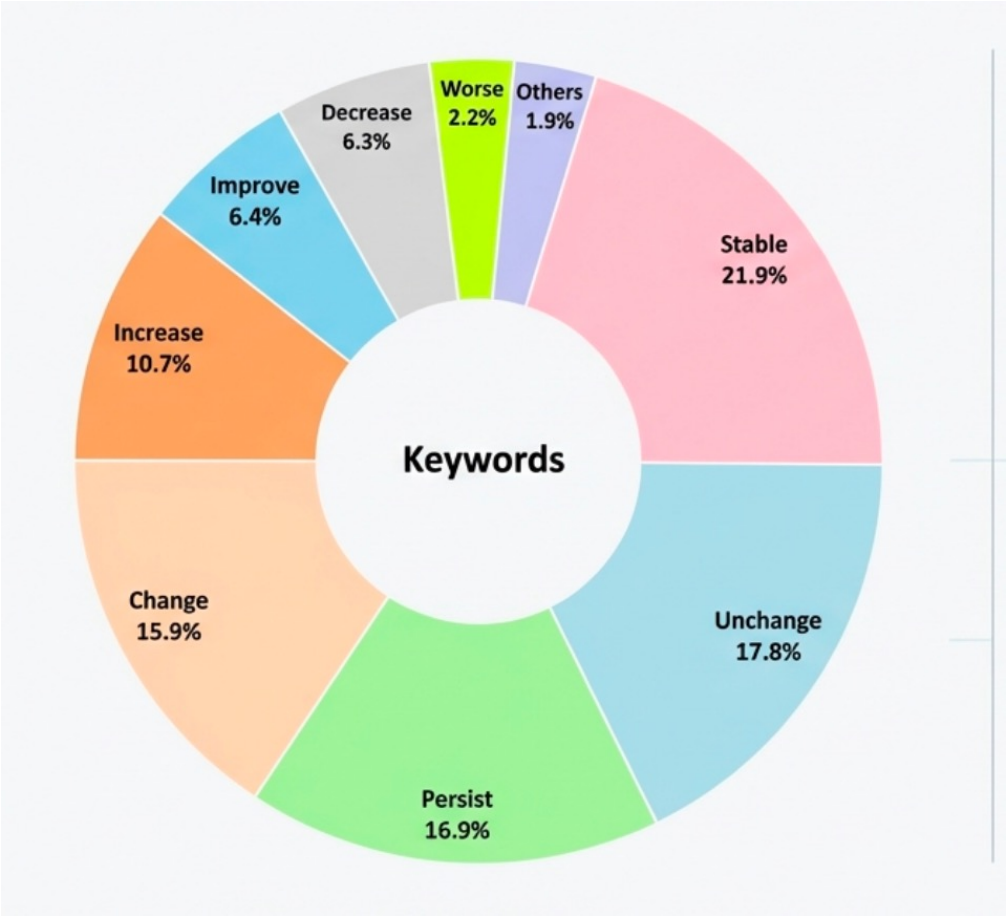

Image Source:Gemini

Figure 3: Vocabulary

The Vocabulary of Progress

Standard datasets focus on static pathology terms (e.g., ”pneumonia"). ’TEMMED-BENCH filters for temporal keywords that describe the *delta*.

"*mild* atelectasis", "*moderate* atelectasis", or "*severe* atelectasis", **Single Visit description**

"*persistent* atelectasis", "*worsening* atelectasis", or "atelectasis has *improved*", where the adjectives indicate a change in condition, we refer to it as a *condition change description sentence*.

18,144 total instances where text is grounded in the difference between two time points.

G

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Dataset Construction Pipeline

Source: CheXpert Plus dataset (224,316 chest radiographs)

1

Condition-Change Report Selection

- Define keywords (increase, worsen, stable)
- Use regex to extract change sentences
- Retain reports where ALL sentences describe changes

2

Current Image Selection

- Retrieve frontal-view image for each target report

3

Historical Image Matching

- Use patient_report_date_order to track prior visits
- Select most recent previous visit as historical reference

i. Identify reports with change-based syntax (e.g,, "worsening", “resolving”).

2. Retrieve corresponding current images.

3. Trace patient ID to retrieve the immediate historical visit image.

✓ Final Dataset: 18,144 instances of (historical image, current image, condition-change report)

VQA Task Construction

Input: Report Sentences

Example Report:
1. Lines and tubes unchanged.
2. Stable pulmonary edema.
3. Decreased aeration of right lung.
4. Increased right pleural effusion.

Output: Question-Answer Pairs

Generated by GPT-4o:
Q: Were lines/tubes changed? A: No
Q: Is pulmonary edema stable? A: Yes
Q: Has right lung aeration increased? A: No
Q: Has pleural effusion decreased? A: No

Quality Control:
- • Balanced 50/50 Yes/No distribution
- • Manual review of all questions
- • Fixed ~11% of AI-generated questions to ensure temporal focus

Prompts for VQA(Closed Book)

Prompts for VQA Task (Closed-book)

Last visit image: <image>

Current visit image: <image>

You are a professional radiologist. You are provided with two X-ray images from the same patient. The first image is from the last visit, and the second image is from the current visit.

I will ask you a question about the change in condition between the last and current visit of this patient. Please answer the question based on the two images and choose from the following two options: [Yes, No]. Please only include your final choice of 'Yes' or 'No' in your response.

Question: {VQA_QUESTION}

Figure 4: Prompts for VQA

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Image-Pair Selection Task Construction

1

Step 1: Input (The Raw Report)

Report: "Comparison to prior study shows **improving mild cardiogenic edema** in the lower lobes."

2

Step 2: Pattern Matching

System scans sentence with RegEx:

- ✓ Keyword: "Improving" (Change Keyword)
- ✓ Middle: "mild cardiogenic" (0-4 attributives)
- ✓ Keyword: "edema" (Pathology Keyword)

3

Step 3: Extraction

System extracts the specific phrase: ***"improving mild cardiogenic edema"***

This extracted phrase becomes the medical statement for the model to match

Pattern Matching: Keywords

Regular Expression Pattern:

[Change Keyword] + 0-4 attributives + [Pathology Keyword]

Change Keywords (KWc)

stable • unchanged • change
persist • increase • decrease
improve • worsen
thicken • thin
progress • deteriorate
reduce • resolve • exacerbate

Pathology Keywords (KWp)

atelectasis • cardiomegaly
consolidation • edema
fracture • lesion • opacity
effusion • pneumonia
pneumothorax

Example: "improving" + "mild cardiogenic" + "edema" = Match ✓

Extraction Phase: Creating the Question

Extracted Medical Statement:

"improving mild cardiogenic edema"

The Question Posed to the Model:

"Select the image pair that best matches: 'improving mild cardiogenic edema'"

Options Structure (A / B / C):

✓ Correct Option	✗ Distractor 1	✗ Distractor 2
Actual image pair from that report	Same pathology (edema) DIFFERENT change	Same pathology (edema) DIFFERENT change

Experimental Setup

Proprietary Models (6)

- GPT o3
- GPT o4-mini
- GPT 4.1
- GPT 4o
- Claude 3.5 Sonnet
- Gemini 2.5 Flash

Open-Source Models (6)

- LLaVA-Med (7B) *
- HuatuoGPT-Vision (7B) *
- HealthGPT (14B) *
- Qwen2.5-VL (7B)
- Llama3.2-Vision (11B)
- LLaVA-OneVision (7B)

** Medical LVLMs*

Evaluation Metrics:

VQA & Selection: Accuracy, F1 | Report Generation: BLEU, ROUGE-L, METEOR

Eval metrics

- **VQA & Image-Pair Selection Metrics**

  - **Accuracy:** Measures the raw percentage of correct predictions (Yes/No for VQA, A/B/C for Selection).
  - **F1 Score (VQA):** A critical metric that balances **precision** and **recall** into a single value, ensuring the model isn't just guessing the most common answer (e.g., always saying "Yes").

- **Report Generation Metrics**
  - **BLEU:** Evaluates the precision of the generated report by measuring exact word overlap against the doctor's reference report.
  - **ROUGE-L:** Focuses on **recall** to ensure the model captures all key diagnostic information and structural flow found in the human reference.
  - **METEOR:** A more advanced metric that accounts for **synonyms and linguistic variations** (e.g., recognizing that "worsening" and "deteriorating" mean the same thing), aligning closer to human judgment.

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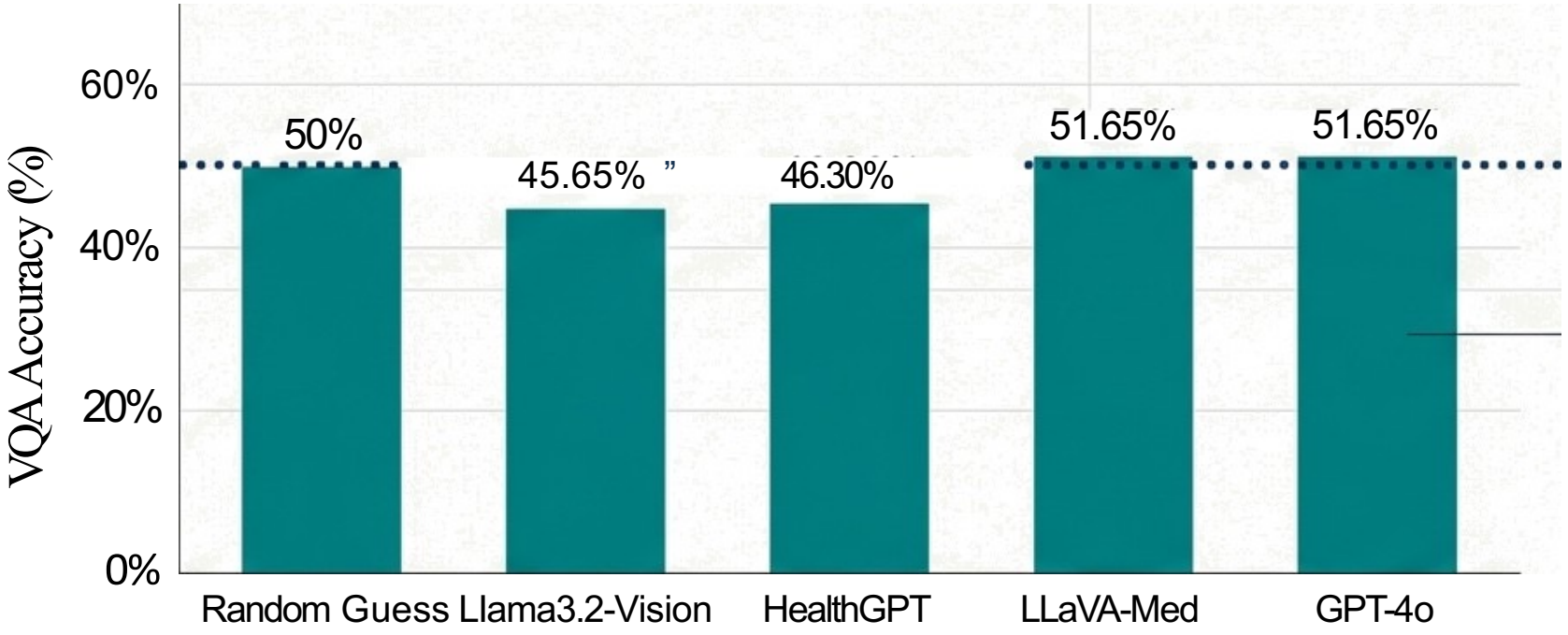
The Closed-Book Failure

Without external knowledge, models struggle to see time.



Random Baseline

Most models perform near or below the random guessing baseline.

Without access to patient history or specific temporal training, even state-of-the-art models lack the innate ability to reason about progression.

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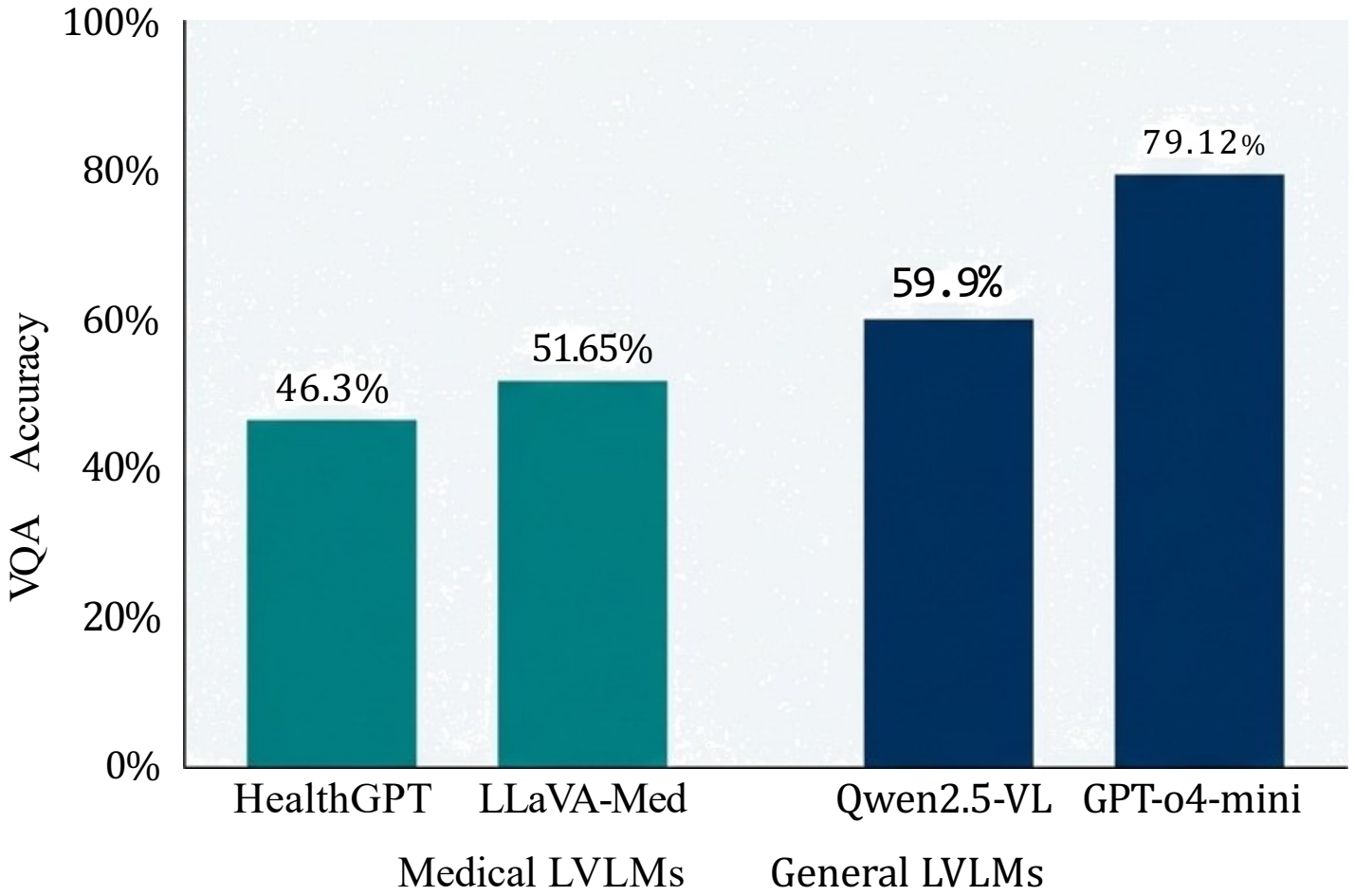
The 'Medical Model' Paradox.



Insight: Catastrophic Forgetting

Expectation: Medical models should win on X-rays.

Reality: Domain fine-tuning often erodes general reasoning capabilities ('Catastrophic Forgetting').
Temporal analysis requires logic, not just vocabulary.

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Model	Params	VQA		Report Generation				Image Selection
		Acc [50]	F1 [50]	BLEU	ROUGE-L	METEOR	Avg.	Acc [33.3]
Open-Source LVLMs								
LLaVA-Med *	7B	51.65	35.23	9.85	6.51	7.10	7.82	/
HuatuoGPT-Vision *	7B	53.00	41.65	7.00	6.54	18.30	10.61	33.29
HealthGPT *	14B	46.30	43.49	11.61	9.26	18.70	13.19	33.64
Qwen2.5-VL	7B	59.90	57.60	12.13	10.46	18.34	13.64	33.87
Llama3.2-Vision	11B	45.65	45.48	8.57	7.94	15.77	10.76	33.06
LLaVA-OneVision	7B	63.90	62.12	5.18	6.07	13.10	8.12	32.83
Proprietary LVLMs								
Gemini 2.5 Flash	/	47.30	47.30	20.23	14.19	22.79	19.07	39.33
Claude 3.5 Sonnet	/	69.90	69.49	17.17	14.01	24.91	18.70	33.53
GPT 4o	/	51.65	47.16	12.74	12.32	23.46	16.17	32.60
GPT 4.1	/	57.90	57.51	9.81	11.67	22.6	14.69	35.38
GPT o4-mini	/	79.15	78.94	20.54	15.75	25.71	20.67	35.03
GPT o3	/	64.40	64.40	16.99	13.71	25.77	18.82	38.05

Table 4: Evaluation results on TEMMED-BENCH with text-only and multi-modal retrieval augmentation (using top-1 retrieval). The highest and second-highest scores for each model in each task are highlighted in red and blue , respectively. Relative performance changes compared to the closed-book setting are shown as subscripts, with red indicating gains and blue indicating drops.

Multi-Modal Retrieval Augmented Generation

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Multi-Modal RAG: Problem Formulation

Text-Only RAG (Previous Work)

Query: (ih, ic, q)

↓

Retriever R returns:

T = [t1, t2, ..., tN]

(Text reports only)

↓

LVLM input:

(ih, ic, q, T)

Multi-Modal RAG (Our Work)

Query: (ih, ic, q)

↓

Retriever R returns:

(Ih, Ic, T)

(Images + Reports)

↓

LVLM input:

(ih, ic, q, Ih, Ic, T)

💡 **Multi-modal retrieval provides richer clinical context with visual evidence from similar cases**

Pairwise Image Retrieval Method

Why Standard Methods Fail:

❌ Image-to-Text: Report describes CHANGE between two images, not a single image
❌ Image-to-Image: Ignores starting point - same current state can mean opposite changes

Pairwise Image Retrieval:

$$\text{Score} = \text{Sim}(\text{Enci}(i_h), \text{Enci}(i^*_h)) + \text{Sim}(\text{Enci}(i_c), \text{Enci}(i^*_c))$$

- First term: Ensures historical images are similar
- Second term: Ensures current images are similar
- Joint evaluation: Ensures condition CHANGES are similar

✓ Only by matching BOTH time points can we find patients who 'walked the same path'

Open Source Models

Retrieval augmentation yields large VQA gains, especially with multi-modal retrieval.

HealthGPT, Llama3.2-Vision, and LLaVA-OneVision show major VQA improvements (+23.60%, +18.45%, and +14.75%)

Multi-modal retrieval is more effective than text-only retrieval, indicating strong dependence on visual context.

Report generation and image-pair selection remain weaker, reflecting limitations in language organization and multi-image reasoning.

Model	VQA		Report Generation				Image Selection
	Acc	F1	BLEU	ROUGE-L	METEOR	Avg.	Acc
Open-Source LVLMs							
LLaVA-Med *	51.65	35.23	9.85	6.51	7.10	7.82	/
+ Text-only RAG	$54.45_{+2.80}$	$41.71_{+6.48}$	17.51	13.35	17.76	$16.21_{+8.39}$	/
+ Multi-Modal RAG	$56.00_{+4.35}$	$44.70_{+9.47}$	12.70	10.74	17.16	$13.53_{+5.71}$	/
HuatuoGPT-Vision *	53.00	41.65	7.00	6.54	18.30	10.61	33.29
+ Text-only RAG	$59.50_{+6.50}$	$52.50_{+10.85}$	8.12	7.53	20.63	$12.09_{+1.48}$	-
+ Multi-Modal RAG	$61.75_{+8.75}$	$56.73_{+15.08}$	9.50	9.07	21.81	$13.46_{+2.85}$	$32.02_{-1.27}$
HealthGPT *	46.30	43.49	11.61	9.26	18.70	13.19	33.64
+ Text-only RAG	$59.05_{+12.75}$	$57.13_{+13.64}$	13.46	10.50	20.68	$14.88_{+1.69}$	-
+ Multi-Modal RAG	$69.90_{+23.60}$	$68.71_{+25.22}$	14.96	11.46	20.42	$15.61_{+2.42}$	$33.53_{-0.11}$
Qwen2.5-VL	59.90	57.60	12.13	10.46	18.34	13.64	33.87
+ Text-only RAG	$63.15_{+3.25}$	$60.26_{+2.66}$	12.80	12.33	22.12	$15.75_{+2.11}$	-
+ Multi-Modal RAG	$65.35_{+5.45}$	$63.03_{+5.43}$	13.26	12.43	21.69	$15.79_{+2.15}$	$35.15_{+1.28}$
Llama3.2-Vision	45.65	45.48	8.57	7.94	15.77	10.76	33.06
+ Text-only RAG	$63.05_{+17.40}$	$58.76_{+13.28}$	12.46	10.86	20.13	$14.48_{+3.72}$	-
+ Multi-Modal RAG	$64.10_{+18.45}$	$60.52_{+15.04}$	14.26	12.37	21.45	$16.03_{+5.27}$	$35.15_{+2.09}$
LLaVA-OneVision	63.90	62.12	5.18	6.07	13.10	8.12	32.83
+ Text-only RAG	$78.25_{+14.35}$	$78.21_{+16.09}$	9.30	9.12	19.81	$12.74_{+4.63}$	-
+ Multi-Modal RAG	$78.65_{+14.75}$	$78.35_{+16.23}$	11.36	10.15	20.27	$13.93_{+5.81}$	$33.64_{+0.81}$

Table 5: Open source models evaluation results

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Proprietary Model results

- **Multi-modal retrieval consistently outperforms text-only retrieval** for all proprietary models.
- **Claude 3.5 Sonnet and GPT-4o show notable VQA gains** (+7.90% and +4.75%) under multi-modal retrieval.
- **Report generation improves substantially**, especially for GPT-4o (+1.59 average score), indicating strong language synthesis benefits.
- **Limited gains in image-pair selection**, suggesting difficulty in leveraging retrieved information across multiple image pairs.
- **Overall, proprietary LVLMs remain strongest in language-heavy and multi-image tasks**, even with modest retrieval-induced gains.

Proprietary LVLMs							
Gemini 2.5 Flash	47.30	47.30	20.23	14.19	22.79	19.07	39.33
+ Text-only RAG	$49.95_{+2.65}$	$49.20_{+1.90}$	22.88	17.27	22.98	$21.04_{+1.97}$	-
+ Multi-Modal RAG	$51.40_{+4.10}$	$50.54_{+3.24}$	23.49	21.48	23.17	$22.71_{+3.64}$	$40.26_{+0.93}$
Claude 3.5 Sonnet	69.90	69.49	17.17	14.01	24.91	18.70	33.53
+ Text-only RAG	$66.25_{-3.65}$	$65.50_{-3.99}$	20.58	16.73	27.27	$21.53_{+2.83}$	-
+ Multi-Modal RAG	$74.15_{+4.25}$	$73.95_{+4.46}$	22.35	17.83	26.44	$22.21_{+3.51}$	$37.35_{+3.82}$
GPT 4o	51.65	47.16	12.74	12.32	23.46	16.17	32.60
+ Text-only RAG	$60.10_{+8.45}$	$59.57_{+12.41}$	21.06	17.17	26.74	$21.66_{+5.48}$	-
+ Multi-Modal RAG	$64.85_{+13.20}$	$64.42_{+17.26}$	23.79	19.14	26.80	$23.24_{+7.07}$	$34.69_{+2.09}$
GPT 4.1	57.90	57.51	9.81	11.67	22.6	14.69	35.38
+ Text-only RAG	$58.50_{+0.60}$	$58.30_{+0.79}$	17.07	15.57	27.48	$20.04_{+5.35}$	-
+ Multi-Modal RAG	$58.60_{+0.70}$	$58.55_{+1.04}$	19.03	16.56	28.07	$21.22_{+6.53}$	$33.87_{-1.51}$
GPT o4-mini	79.15	78.94	20.54	15.75	25.71	20.67	35.03
+ Text-only RAG	$81.80_{+2.65}$	$81.78_{+2.84}$	24.77	19.42	27.99	$24.06_{+3.39}$	-
+ Multi-Modal RAG	$77.80_{-1.35}$	$77.69_{-1.25}$	24.30	19.06	26.46	$23.27_{+2.61}$	$34.69_{-0.34}$
GPT o3	64.40	64.40	16.99	13.71	25.77	18.82	38.05
+ Text-only RAG	$66.65_{+2.25}$	$66.60_{+2.20}$	18.89	15.33	26.84	$20.35_{+1.53}$	-
+ Multi-Modal RAG	$65.65_{+1.25}$	$65.60_{+1.20}$	18.72	15.20	27.16	$20.44_{+1.61}$	$39.56_{+1.51}$

Table 6: Proprietary source models evaluation results

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Ablation Study & Analysis

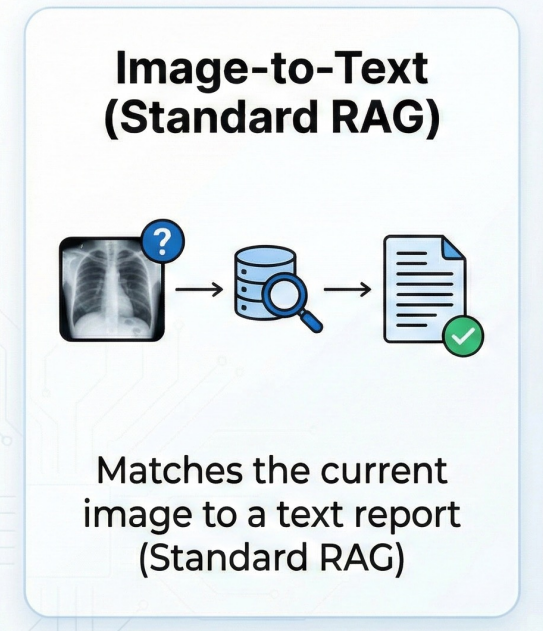

Image Source:Gemini

The AI looks at the patient's **Current Image** and searches the database for a **Report** that matches it mathematically.

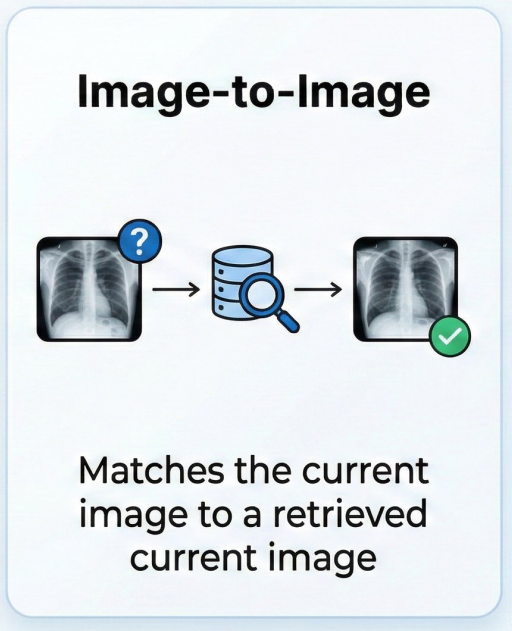

Image Source:Gemini

The AI looks at the patient's **Current Image** and searches the database for another **Current Image** that looks similar.

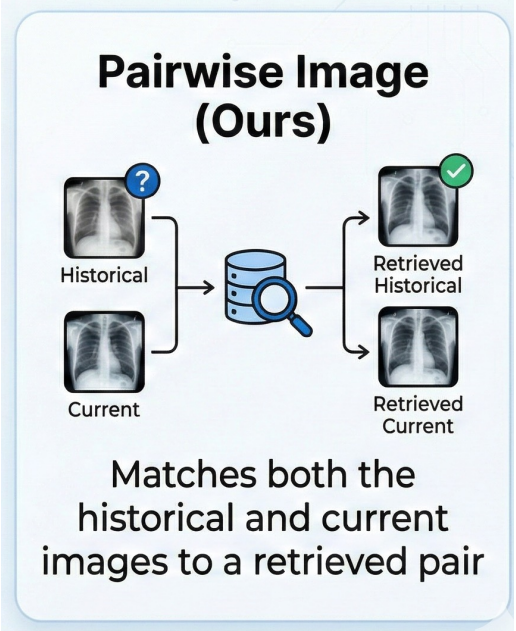

Image Source:Gemini

The AI looks at the patient's **Current Image and Historical** and searches the database for another **Current and Historical Image** that looks similar.

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Results

- **Multi-modal RAG outperforms text-only RAG** across all retrieval methods in both accuracy and F1-score.

- For **Image-to-Text retrieval**, accuracy improves from **58.90 → 65.75**, and F1-score from **56.77 → 64.19**.

- For **Image-to-Image retrieval**, accuracy increases from **58.15 → 68.45**, and F1-score from **55.95 → 67.17**.

- **Pairwise Image retrieval** shows the largest gain, with accuracy improving from **59.05 → 69.90** and F1-score from **57.13 → 68.71**.

Retrieval Method	Acc	F1
Text-only RAG		
Image-to-Text	58.90	56.77
Image-to-Image	58.15	55.95
Pairwise Image	**59.05**	**57.13**
Multi-Modal RAG		
Image-to-Text	65.75	64.19
Image-to-Image	68.45	67.17
Pairwise Image	**69.90**	**68.71**

Table 7: Evaluation of of HealthGPT with various retrieval methods

https://doi.org/10.48550/arXiv.2509.25143

Top-k Retrieval Analysis

Impact of Number of Retrieved Examples

Finding: More retrieved examples → Better performance

HealthGPT:

Top-1: 69.90% → Top-5: 72.27% (+2.37%)

GPT-4o:

Top-1: 64.85% → Top-5: 71.45% (+6.60%)

Insight:

More advanced models benefit MORE from multiple examplesBetter at synthesizing information from multiple retrieved cases

Multi-modal RAG consistently outperforms text-only RAG across all top-k settings

TEMMED-BENCH Robustness: Top-1 Retrieval Hack Test

What is the 'Top-1 Retrieval Hack'?

Find the most similar case in the database → Copy its answer directly (no reasoning)

Results: TEMMED-BENCH vs. Previous Benchmarks

Benchmark	Model + RAG	Top-1 Hack	Winner
IU-Xray	29.80	30.03	Hack Wins 😱
MIMIC-CXR	18.69	23.81	Hack Wins 😱
TEMMED-BENCH	26.41	23.50	Model Wins ✓

💡 **TEMMED-BENCH requires reasoning, not just pattern matching → More robust benchmark**

Key Takeaways

TEMMED-BENCH fills a critical gap

First benchmark for temporal medical image reasoning

Current LVLMs struggle significantly

Most models perform at random-guessing level. Only reasoning models (GPT o4-mini: 79.15%) show promise.

Multi-modal RAG is highly effective

Average VQA improvement of +2.59%. Some models gain +23% with visual context.

Pairwise retrieval matters

Matching both historical and current images ensures similar condition trajectories.

Medical models need better training

Domain fine-tuning erodes reasoning. Need frameworks that preserve general capabilities.

→ TEMMED-BENCH provides an evaluation grounded in real clinical practice

Discussion

- What else can be done to increase the effectiveness of the medical LVLMS?

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Thank You for Listening